Part A – Introduction to Random Networks

1. **Basic Erdős–Rényi Model**
   - Generate an Erdős–Rényi random graph $G(n,p)$ with:
     - $n = 100$ nodes
     - edge probability $p = 0.05$.
   - **Questions:**
     a. How many edges are present in your generated graph?
     b. What is the average degree of the network?
     c. Does the degree distribution look Poisson-like? Plot and comment.

Part B – Connectivity & Components

2. **Connected Components**
   - For $G(100,0.01)$:
     a. Count the number of connected components.
     b. Identify the size of the largest connected component.
     c. Plot the distribution of component sizes.
3. **Threshold Phenomena**
   - For increasing values of p (e.g., $p = 0.01, 0.05, 0.1, 0.2$):
     a. Estimate the probability that the graph is connected.
     b. At which approximate p does the graph "almost surely" become connected?

Part C – Clustering & Path Length

4. **Clustering Coefficient**
   - Compare the average clustering coefficient of:
     a. A random graph $G(200,0.05)$.
     b. A Watts–Strogatz small-world network with the same number of nodes and average degree.
   - What do you observe?
5. **Average Path Length**
   - Compute the average shortest path length for G(200,0.05).
   - Compare with the Watts–Strogatz model.
   - Discuss: Do random graphs exhibit the "small-world" property?

Part D – Scaling & Degree Distributions

6. **Degree Distribution Scaling**
   - For n=1000, $p = 0.01$:
     a. Plot the degree distribution.
     b. Compare it with a binomial distribution Binomial($n-1$, p).

c. Discuss: Why does it resemble a Poisson distribution when n is large and p small?

7. **Power-law vs Random Networks**
    - Generate a Barabási–Albert (BA) preferential attachment graph with
    - n=1000,m=3.
    - Compare its degree distribution with that of an Erdős–Rényi random graph with similar average degree.
    - **Discussion:** What fundamental difference do you notice?